

Long chain omega-3 fatty acids intake, fish consumption and mental disorders in the SUN cohort study.

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BACKGROUND: Very long chain omega-3 fatty acids (w-3 PUFA) intake and fish consumption have been suggested as protective factors against neuropsychiatric disorders but there is scarcity of large cohort studies assessing this association. AIM OF THE STUDY: To assess the association between w-3-PUFA intake and fish consumption and mental disorders. METHODS: A prospective cohort study was performed in 7,903 participants. W-3 PUFA intake and fish consumption were ascertained through a validated semi-quantitative food frequency questionnaire. The outcomes after 2 years of follow-up were: (1) Incident mental disorder (depression, anxiety, or stress), (2) incident depression, and (3) incident anxiety. Logistic regression models and generalized additive models were fit to assess the relationship between w-3 PUFA intake or fish consumption and the incidence of these outcomes. Odds ratios (OR) and their 95% confidence intervals (CI) were calculated. RESULTS: 173 cases of depression, 335 cases of anxiety, and 4 cases of stress were observed during 2-year follow-up. ORs (95% CI) of mental disorder for successive quintiles of energy-adjusted w-3 PUFA intake were 1 (reference), 0.72 (0.52-0.99), 0.79 (0.58-1.08), 0.65 (0.47-0.90), and 1.04 (0.78-1.40). Subjects with a moderate consumption of fish (third and fourth quintiles of consumption: median of each quintile 83.3 and 112 g/day, respectively) had a relative risk reduction higher than 30%. CONCLUSIONS: A potential benefit of w-3 PUFA intake on total mental disorders is suggested, although no linear trend was apparent.